

SCHOOL
OF
INFORMATION AND COMMUNICATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAMME STRUCTURE

M.TECH. COMPUTER SCIENCE AND ENGINEERING

SPECIALIZATION : DATA SCIENCE

Academic Batch 2026-2028 and Onwards

GAUTAM BUDDHA UNIVERSITY

GAUTAM BUDH NAGAR, GREATER NOIDA, UP, INDIA

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SEMESTER I

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CS521	Advanced Data Base Management System	3	0	1	3	PCC
2	CS523	Advanced Data Structure and Algorithm	3	1	0	4	PCC
3	CS525	Research Techniques in IT	3	0	0	3	PCC
4	CS527	Python Programming	3	0	0	3	PCC
5	ES415	Energy and Environment	3	0	0	3	PCC
6	CD581	Advanced Data Base Management System Lab	0	0	4	2	PCC
7	CD583	Python Programming Lab	0	0	4	2	PCC
Total Hours and Credits			15	1	4	20	

SEMESTER II

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CD522	Statistical Foundations for Data Science	3	0	0	3	PCC
2	CD524	Deep Learning	3	0	0	3	PCC
3	CD526	ML Ops	3	0	0	3	PCC
4	CS528	Blockchain Technology	3	1	0	3	PCC
5	CSXXX	Programme Elective-I	3	0	0	3	PCC
6	CD582	Statistical Foundations for Data Science Lab	0	0	4	2	PCC
7	CS584	Deep Learning Lab	0	0	4	2	PCC
Total Hours and Credits			15	1	4	20	

ELECTIVES FROM DCSE

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CD530	Natural Language Processing	3	0	0	3	E1
2	CD532	Data Acquisition and Production	3	0	0	3	
3	CD534	Predictive Analytics and Data Visualization	3	0	0	3	
4	CD536	Data Science for Business Analytics	3	0	0	3	
5	CS538	Parallel and High Performance Computing	3	0	0	3	

SEMESTER III

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CS621	Cloud and Edge Computing	3	0	0	3	PCC
2	CD623	Big Data Analytics	3	0	0	3	PCC
3	CDXXX	Programme Elective-II	3	0	0	3	PCC
4	CDXXX	Programme Elective-III	3	0	0	3	PCC
5	CD681	Big Data Analytics Lab	0	0	4	2	PCC
6	CD691	Dissertation Part-I	0	0	16	20	Dissertation
Total Hours and Credits			12	0	16	20	

SEMESTER IV

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CD692	Dissertation Part-II	0	0	40	20	Dissertation
Total Hours and Credits			0	0	40	20	

ELECTIVE BASKET-III/III (SELECT ANY TWO)

S.No.	Course Code	Course Name	L	T	P	Credits	Types
1	CD631	Data Mining and Knowledge Discovery	3	0	0	3	E2
2	CD633	Time Series Analysis and Forecasting	3	0	0	3	E2
3	CD635	Agentic AI	3	0	0	3	E2
4	CD637	Data Engineering and Modern Data Warehousing	3	0	0	3	E2
5	CD639	Digital Transformation using AI	3	0	0	3	E2
6	CD641	Causal Inference and Experimentation	3	0	0	3	E2
7	CD643	Graph Machine Learning and Network Analytics	3	0	0	3	E3
8	CD645	Responsible and Explainable AI	3	0	0	3	E3
9	CD647	Optimization Methods for Data Science	3	0	0	3	E3
10	CD649	Data Security	3	0	0	3	E3





